

# Mereotopological Ontologies with Compositional Algebras

A Formalization in Lean 4

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2025-2

## **Abstract**

We develop the mathematical foundations for formalizing mereotopological ontologies equipped with compositional algebras that function as deductive systems. The key insight is that Heyting algebras—the algebraic semantics for intuitionistic logic—provide the natural framework for such compositional structures. We present a complete, self-contained Lean 4 formalization.

## **Contents**

# 1 Introduction

Given a formal ontology  $\mathcal{O}$  with mereotopological structure (part-whole relations and topological connection) and an algebra  $\mathcal{A}$  acting on  $\mathcal{O}$ , we seek to formalize:

1. The mereotopological structure of  $\mathcal{O}$
2. The algebraic structure of  $\mathcal{A}$  as a Heyting algebra
3. The action  $[\![\cdot]\!] : \mathcal{O} \rightarrow \mathcal{A}$  preserving relevant structure
4. The deductive character emerging from the Heyting implication

# 2 Mereotopological Ontology

**Definition 2.1** (Mereotopological Ontology). A *mereotopological ontology* is a structure  $\mathcal{O} = (O, \sqsubseteq, C)$  where:

- $O$  is a non-empty set of *concepts* or *entities*
- $\sqsubseteq$  is a *parthood* relation (partial order)
- $C$  is a *connection* relation (reflexive, symmetric)

satisfying the *mereotopological coherence axiom*:

$$x \sqsubseteq y \implies C(x, y)$$

The parthood relation  $\sqsubseteq$  satisfies:

- **Reflexivity:**  $\forall x. x \sqsubseteq x$
- **Transitivity:**  $x \sqsubseteq y \wedge y \sqsubseteq z \implies x \sqsubseteq z$
- **Antisymmetry:**  $x \sqsubseteq y \wedge y \sqsubseteq x \implies x = y$

**Definition 2.2** (Derived Mereotopological Relations). From the primitive relations, we define:

$$\begin{array}{ll} \text{PP}(x, y) := x \sqsubseteq y \wedge x \neq y & \text{(proper parthood)} \\ \text{O}(x, y) := \exists z. z \sqsubseteq x \wedge z \sqsubseteq y & \text{(overlap)} \\ \text{D}(x, y) := \neg \text{O}(x, y) & \text{(disjointness)} \\ \text{EC}(x, y) := C(x, y) \wedge \text{D}(x, y) & \text{(external connection)} \end{array}$$

*Remark 2.3.* The connection relation  $C$  captures topological adjacency without requiring shared parts. Two concepts can be externally connected (touching) without overlapping—a crucial distinction for spatial and conceptual reasoning.

### 3 Heyting Algebras as Compositional Algebras

Recall that a Heyting algebra provides algebraic semantics for intuitionistic logic.

**Definition 3.1** (Heyting Algebra). A *Heyting algebra* is a bounded distributive lattice  $\mathcal{H} = (H, \sqcup, \sqcap, \Rightarrow, 0, 1)$  such that for all  $a, b \in H$ , the *relative pseudo-complement*  $a \Rightarrow b$  exists, characterized by the **adjunction property**:

$$c \sqcap a \leq b \iff c \leq a \Rightarrow b$$

**Theorem 3.2** (Completeness). *For intuitionistic propositional logic:*

$$\Gamma \vdash \varphi \iff \Gamma \models \varphi$$

where  $\Gamma \models \varphi$  means: for every Heyting algebra  $\mathcal{H}$  and valuation  $v$ , if  $\mathcal{H}, v \models \Gamma$  then  $\mathcal{H}, v \models \varphi$ .

#### 3.1 Compositional Algebra Structure

We interpret the Heyting algebra operations as conceptual composition operators:

| Operation   | Notation                   | Interpretation                     |
|-------------|----------------------------|------------------------------------|
| Meet        | $a \sqcap b$               | Conceptual conjunction/composition |
| Join        | $a \sqcup b$               | Conceptual disjunction/choice      |
| Implication | $a \Rightarrow b$          | Conceptual entailment              |
| Negation    | $\sim a = a \Rightarrow 0$ | Conceptual complement              |
| Top         | 1                          | Universal concept                  |
| Bottom      | 0                          | Empty/impossible concept           |

**Definition 3.3** (Derivability). In a compositional algebra  $\mathcal{A}$ , we define the *derivability relation*:

$$a \vdash_{\mathcal{A}} b := a \sqcap b = a$$

This is equivalent to  $a \leq b$  in the underlying order.

**Proposition 3.4.** *Derivability forms a preorder:*

1. **Reflexivity:**  $a \vdash_{\mathcal{A}} a$  (since  $a \sqcap a = a$ )
2. **Transitivity:**  $a \vdash_{\mathcal{A}} b$  and  $b \vdash_{\mathcal{A}} c$  implies  $a \vdash_{\mathcal{A}} c$

### 4 The Ontological Algebra

**Definition 4.1** (Ontological Algebra). An *ontological algebra* is a triple  $(\mathcal{O}, \mathcal{A}, \llbracket \cdot \rrbracket)$  where:

- $\mathcal{O} = (O, \sqsubseteq, C)$  is a mereotopological ontology
- $\mathcal{A}$  is a Heyting algebra (compositional algebra)
- $\llbracket \cdot \rrbracket : O \rightarrow \mathcal{A}$  is an *interpretation function*

satisfying:

1. **Monotonicity:**  $c_1 \sqsubseteq c_2 \implies \llbracket c_1 \rrbracket \leq \llbracket c_2 \rrbracket$
2. **Connection non-triviality:**  $\mathbf{C}(c_1, c_2) \implies \llbracket c_1 \rrbracket \sqcap \llbracket c_2 \rrbracket \neq 0$

**Definition 4.2** (Fundamental Operations). Given concepts  $C_1, C_2 \in \mathcal{O}$ , we define:

$$\begin{aligned} C_1 \triangleright C_2 &:= \llbracket C_1 \rrbracket \sqcap \llbracket C_2 \rrbracket && \text{(combination)} \\ C_1 \multimap C_2 &:= \llbracket C_1 \rrbracket \Rightarrow \llbracket C_2 \rrbracket && \text{(conceptual implication)} \end{aligned}$$

## 4.1 Key Theorems

**Theorem 4.3** (Parthood Implies Derivability). *If  $c_1 \sqsubseteq c_2$  in  $\mathcal{O}$ , then:*

$$\llbracket c_1 \rrbracket \sqcap \llbracket c_2 \rrbracket = \llbracket c_1 \rrbracket$$

*That is,  $\llbracket c_1 \rrbracket \vdash_{\mathcal{A}} \llbracket c_2 \rrbracket$ .*

*Proof.* By monotonicity,  $\llbracket c_1 \rrbracket \leq \llbracket c_2 \rrbracket$ . Then:

$$\llbracket c_1 \rrbracket \sqcap \llbracket c_2 \rrbracket \leq \llbracket c_1 \rrbracket \leq \llbracket c_2 \rrbracket$$

so  $\llbracket c_1 \rrbracket \sqcap \llbracket c_2 \rrbracket \leq \llbracket c_1 \rrbracket$ . Conversely,  $\llbracket c_1 \rrbracket \leq \llbracket c_1 \rrbracket \sqcap \llbracket c_2 \rrbracket$  since  $\llbracket c_1 \rrbracket \leq \llbracket c_1 \rrbracket$  and  $\llbracket c_1 \rrbracket \leq \llbracket c_2 \rrbracket$ .  $\square$

**Theorem 4.4** (Overlap Non-Triviality). *If  $\mathbf{O}(c_1, c_2)$  in  $\mathcal{O}$ , then  $c_1 \triangleright c_2 \neq 0$ .*

*Proof.* From  $\mathbf{O}(c_1, c_2)$ , there exists  $z$  with  $z \sqsubseteq c_1$  and  $z \sqsubseteq c_2$ . By monotonicity,  $\llbracket z \rrbracket \leq \llbracket c_1 \rrbracket$  and  $\llbracket z \rrbracket \leq \llbracket c_2 \rrbracket$ , so  $\llbracket z \rrbracket \leq \llbracket c_1 \rrbracket \sqcap \llbracket c_2 \rrbracket$ . By  $\mathbf{C}$ -reflexivity,  $\mathbf{C}(z, z)$  holds, so by connection non-triviality,  $\llbracket z \rrbracket \sqcap \llbracket z \rrbracket = \llbracket z \rrbracket \neq 0$ . Thus  $c_1 \triangleright c_2 \geq \llbracket z \rrbracket \neq 0$ .  $\square$

**Theorem 4.5** (Modus Ponens for Concepts). *For any  $c_1, c_2 \in \mathcal{O}$ :*

$$\llbracket c_1 \rrbracket \sqcap (c_1 \multimap c_2) \leq \llbracket c_2 \rrbracket$$

*Proof.* This is the Heyting algebra property:  $a \sqcap (a \Rightarrow b) \leq b$ .  $\square$

## 5 Connection to Topological Semantics

Open sets of a topological space form a Heyting algebra, providing a canonical model.

**Proposition 5.1** (Topological Heyting Algebra). *Let  $\mathcal{T}$  be a topological space. Then  $\mathcal{H} = (\mathcal{O}(\mathcal{T}), \cup, \cap, \Rightarrow, \emptyset, \mathcal{T})$  is a Heyting algebra where:*

$$\begin{aligned} A \Rightarrow B &= \text{Int}(-A \cup B) \\ \sim A &= \text{Int}(-A) \end{aligned}$$

**Example 5.2** (Peirce's Law Failure). Consider  $\mathcal{H} = \mathcal{O}(\mathbb{R})$  with  $v(p) = \mathbb{R} \setminus \{0\}$  and  $v(q) = \emptyset$ . Then:

$$\llbracket (p \rightarrow q) \rightarrow p \rrbracket_v = \text{Int}(\emptyset \cup (\mathbb{R} \setminus \{0\})) = \mathbb{R} \setminus \{0\}$$

So  $\llbracket ((p \rightarrow q) \rightarrow p) \rightarrow p \rrbracket_v = \mathbb{R} \setminus \{0\} \neq \mathbb{R}$ .

**Ontologically:** Peirce's law fails because we cannot constructively obtain  $p$  merely from knowing that “if  $p$  implied  $q$ , then  $p$  would hold.”

## 6 The Three-Element Example

Consider the three-element chain  $\mathbf{3} = \{0 < a < 1\}$ :

[scale=0.8] [circle, draw, minimum size=6mm] (0) at (0,0) 0; [circle, draw, minimum size=6mm] (a) at (0,1.5) a; [circle, draw, minimum size=6mm] (1) at (0,3) 1; [thick] (0) -- (a) -- (1);

The operations are:

| $\sqcap$ | 0 | a | 1 | $\Rightarrow$ | 0 | a | 1 |
|----------|---|---|---|---------------|---|---|---|
| 0        | 0 | 0 | 0 | 0             | 1 | 1 | 1 |
| a        | 0 | a | a | a             | 0 | 1 | 1 |
| 1        | 0 | a | 1 | 1             | 0 | a | 1 |

*Remark 6.1.* Note that  $\sim a = a \Rightarrow 0 = 0$ , but  $\sim\sim a = 0 \Rightarrow 0 = 1 \neq a$ . This demonstrates the failure of double negation elimination in intuitionistic logic.

## 7 Lean 4 Formalization

The complete formalization is self-contained, building all structures from first principles.

### 7.1 Algebraic Foundations

```
class PartialOrder' (α : Type u) where
  le : α → α → Prop
  le_refl : ∀ a, le a a
  le_trans : {a b c}, le a b → le b c → le a c
  le_antisymm : {a b}, le a b → le b a → a = b

class HeytingAlgebra (α : Type u) extends BoundedLattice
  where
  himp : α → α → Prop
  le_himp_iff : {a b c}, le (inf a b) c ↔ le a (himp b c)
```

### 7.2 Mereotopological Ontology

```
class MereotopologicalOntology (α : Type u) where
  partOf : α → α → Prop
  C : α → α → Prop
  partOf_refl : ∀ x, partOf x x
  partOf_trans : {x y z}, partOf x y → partOf y z → partOf x z
  partOf_antisymm : {x y}, partOf x y → partOf y x → x = y
  C_refl : ∀ x, C x x
  C_symm : {x y}, C x y → C y x
  part_connected : {x y}, partOf x y → C x y
```

### 7.3 The Ontological Algebra

```

class OntologicalAlgebra (O : Type u) (A : Type v)
  [MereotopologicalOntology O] [CompositionalAlgebra A] where
  interp : O → A
  interp_mono : { c c' : O }, partOf c c' → interp c ≤ interp c'
  connected_nontrivial : { c c' : O }, C c c' →
    inf (interp c) (interp c') = bot

```

### 7.4 Key Theorem in Lean

```

theorem partOf_derives { c c' : O }
  (h : MereotopologicalOntology.partOf c c') :
  c ≤ c' := by
  apply le_antisymm
  exact inf_le_left c c'
  exact le_inf (le_refl c) (interp_mono h)

```

## 8 Philosophical Remarks

The connection between mereotopological ontology and Heyting algebras illuminates deep correspondences:

1. **Constructive Ontology:** Just as intuitionistic logic requires constructive evidence for assertions, ontological composition requires explicit witnesses (the overlap relation requires a common part).
2. **Failure of Classical Duality:** The asymmetry between  $\varphi \rightarrow \neg\neg\varphi$  (valid) and  $\neg\neg\varphi \rightarrow \varphi$  (not valid) mirrors the asymmetry in mereotopology: parthood implies connection, but connection does not imply overlap.
3. **Topological Intuition:** Interpreting concepts as open sets provides geometric intuition. A concept “close to being true” (like  $\mathbb{R} \setminus \{0\}$ ) is not the same as being true ( $\mathbb{R}$ )—the boundary matters.
4. **Compositional Semantics:** The adjunction  $a \sqcap b \leq c \Leftrightarrow a \leq b \Rightarrow c$  captures the essence of compositional reasoning.

*Remark 8.1* (BHK Interpretation). Under the BHK interpretation, a construction of  $\varphi_1 \rightarrow \varphi_2$  is a method transforming constructions of  $\varphi_1$  into constructions of  $\varphi_2$ . In our ontological setting:

$$c_1 \multimap c_2 = \llbracket c_1 \rrbracket \Rightarrow \llbracket c_2 \rrbracket$$

represents the “conceptual method” for deriving  $c_2$  from  $c_1$ .

## 9 Conclusion

We have established a rigorous framework connecting:

- **Mereotopology:** Part-whole and connection relations on concepts
- **Heyting Algebras:** The algebraic semantics of intuitionistic logic
- **Compositional Reasoning:** Deductive structure via the implication operation
- **Lean 4 Formalization:** Machine-checked proofs of the core theorems

The Heyting algebra structure naturally captures the intuitionistic character of ontological reasoning: we cannot assert properties of concepts without constructive justification, and the composition of concepts must respect the underlying mereotopological structure.

## References

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3. Heyting, A. *Intuitionism: An Introduction*. North-Holland, 1956.